

DTU



Perception for Autonomous Systems - 34759:

Edge Detection

Lecturer: Evangelos Boukas—PhD

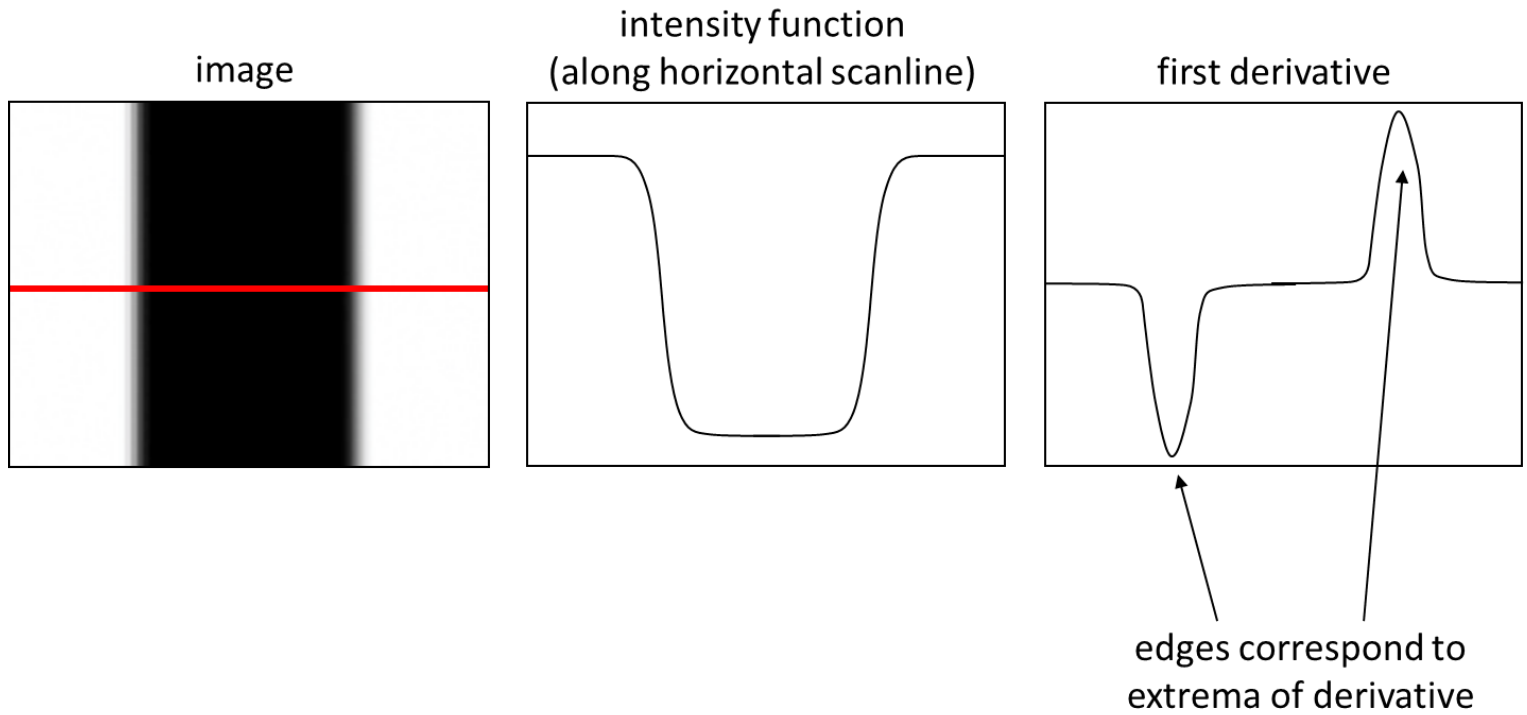
Edge Detection

- What is an Edge?
- Image Derivative
- Gradient
- Sobel
- Laplacian
- Canny



Edge Detection

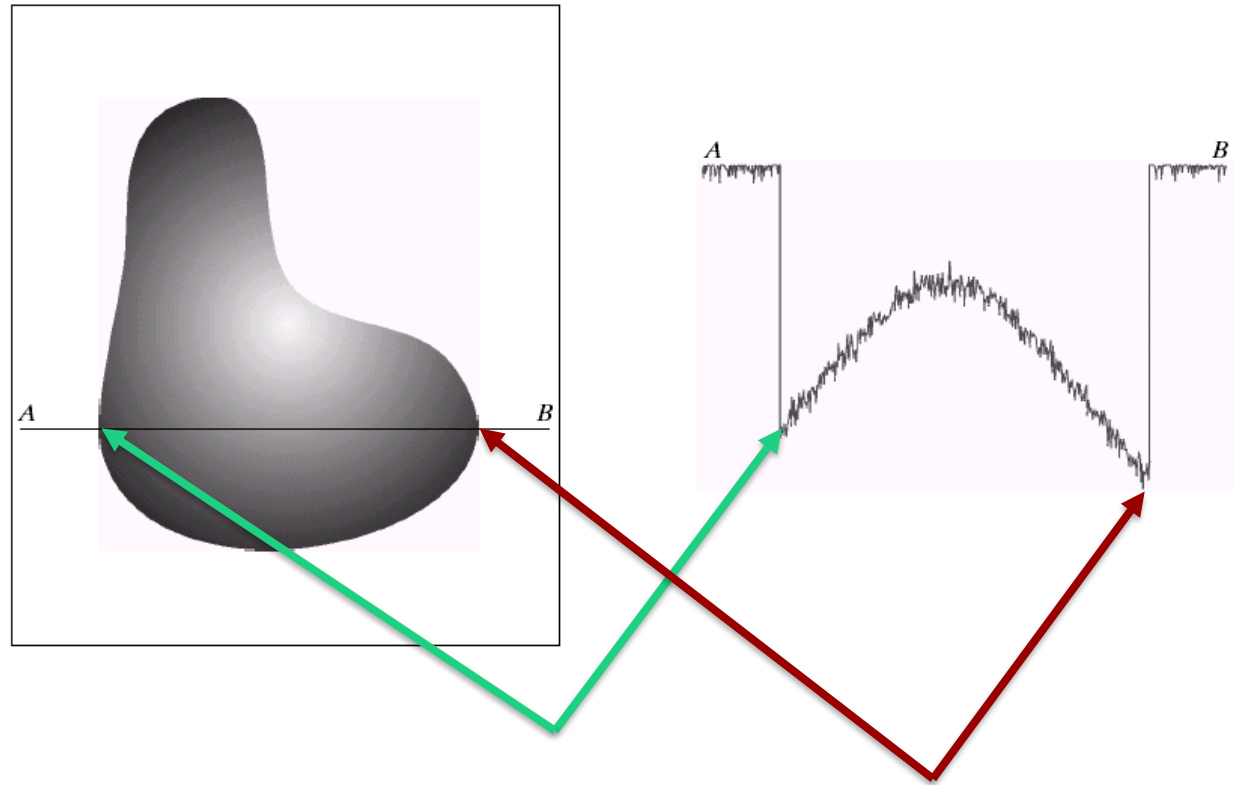
- What is an edge?



Edge Detection

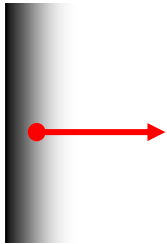
- What is an edge?
- Derivative of an Image:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

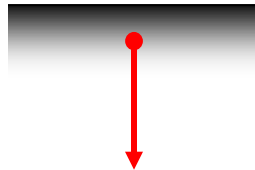


- The gradient is a vector which points in the direction of most rapid change in intensity:

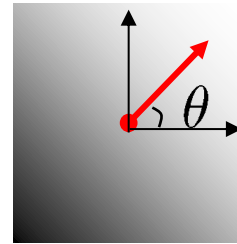
$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right]$$



$$\nabla f = \left[\frac{\partial f}{\partial x}, 0 \right]$$

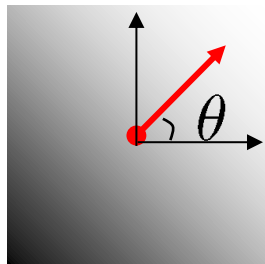


$$\nabla f = \left[0, \frac{\partial f}{\partial y} \right]$$



$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right]$$

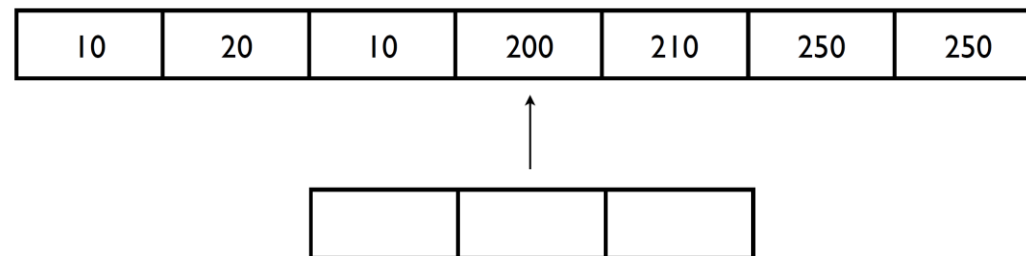
Gradient Simplified



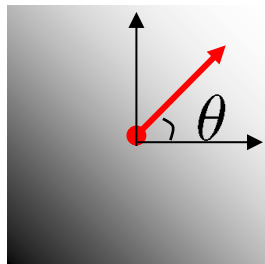
$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right]$$

- How would you define the 1D filter of the gradient:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \longrightarrow \frac{\partial f}{\partial x} = f(x+1, y) - f(x, y)$$



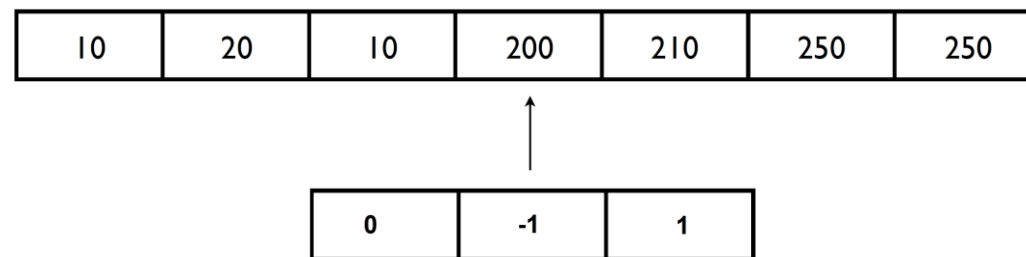
Gradient Simplified



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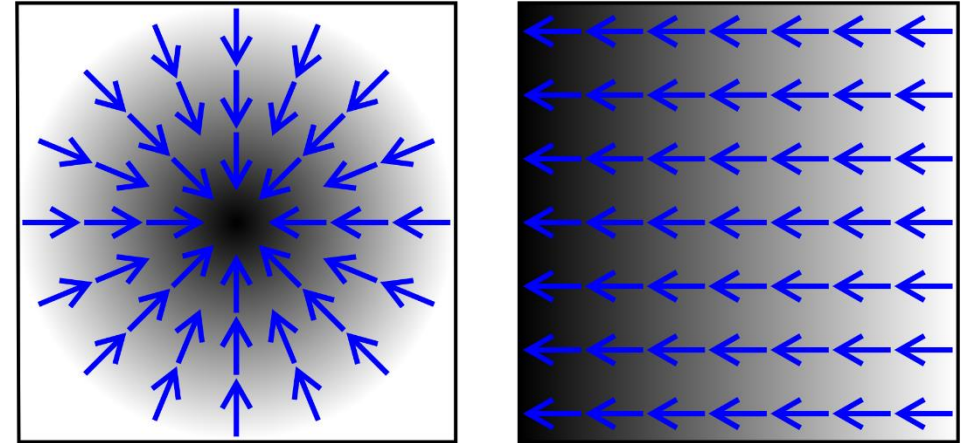
Gradient Simplified

- The gradient is defined by its orientation:

$$\theta = \tan^{-1} \left(\frac{\partial f}{\partial y} / \frac{\partial f}{\partial x} \right)$$

- and magnitude:

$$\|\nabla f\| = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}$$



- Practically:

$$g_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

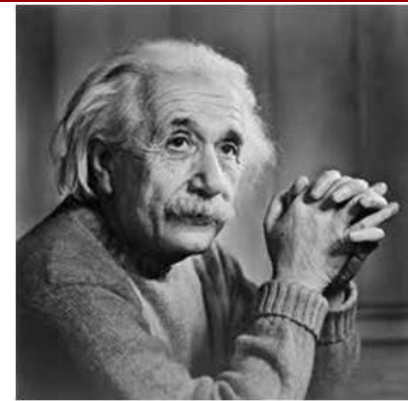
$$g_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

- Magnitude:

$$g = \sqrt{g_x^2 + g_y^2}$$

- Orientation:

$$\Theta = \tan^{-1} \left(\frac{g_y}{g_x} \right)$$



Derivative Filters

Sobel

1	0	-1
2	0	-2
1	0	-1

1	2	1
0	0	0
-1	-2	-1

Scharr

3	0	-3
10	0	-10
3	0	-3

3	10	3
0	0	0
-3	-10	-3

Prewitt

1	0	-1
1	0	-1
1	0	-1

1	1	1
0	0	0
-1	-1	-1

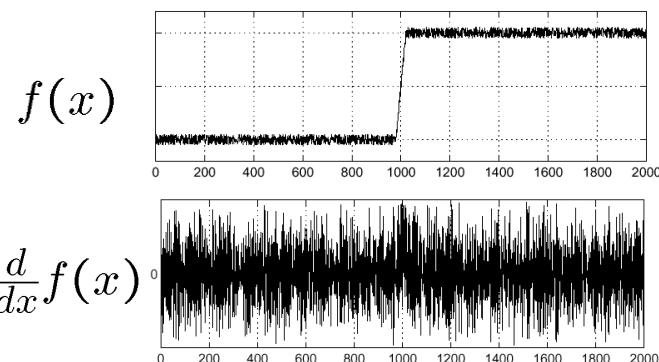
Roberts

0	1
-1	0

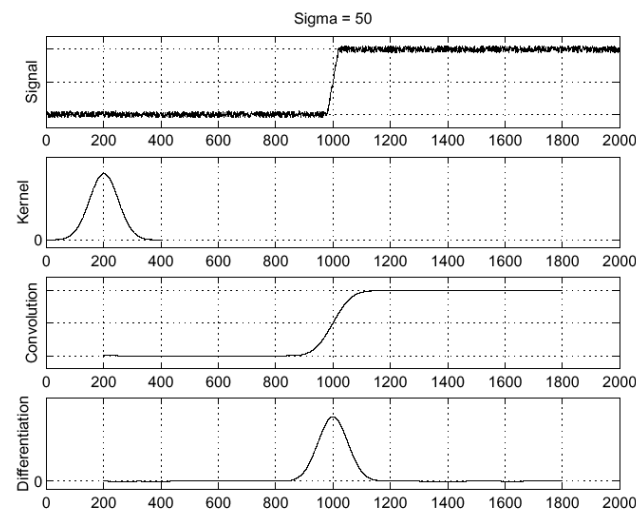
1	0
0	-1

DTU Preprocessing to Edge Detection

- In reality derivatives are very prone to noise
Consider the following example:

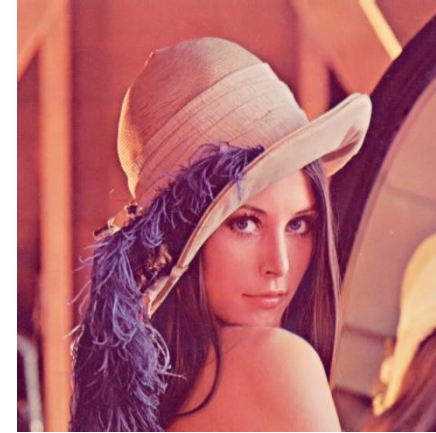
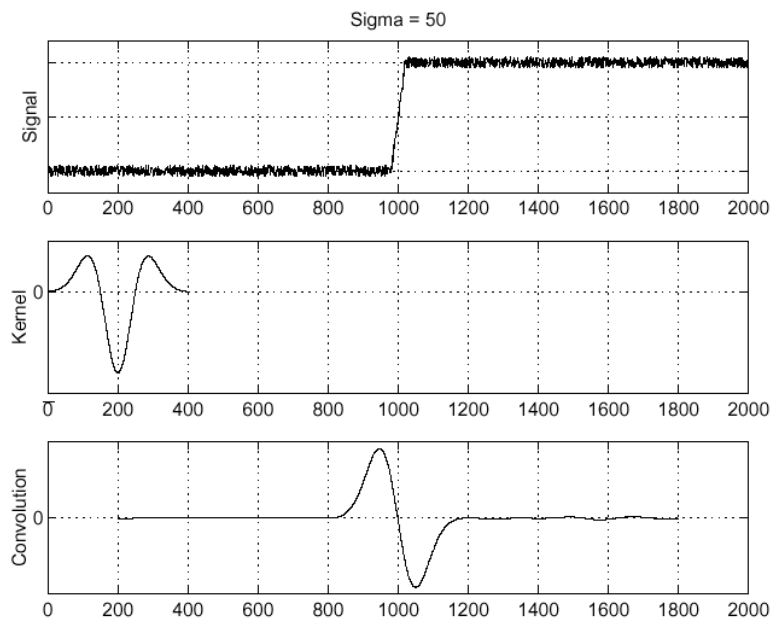


- To overcome this issue, we can smooth the signal beforehand



Laplacian of Gaussian

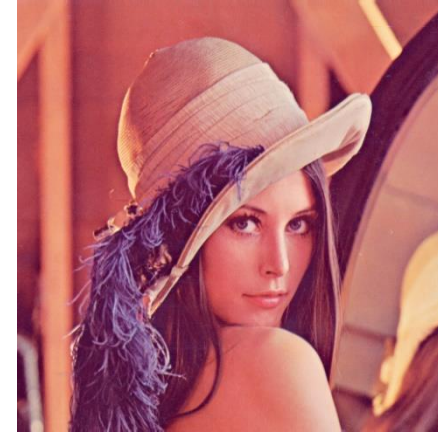
$$\cdot \frac{\partial^2}{\partial x^2}(h \star f)$$



Canny Edge Detection

Example of a Complex system:

- Noise reduction
- Gradient calculation
- Non-maximum suppression
- Double threshold
- Edge Tracking by Hysteresis.



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